A NODAL CURVE OF GENUS AT LEAST 2 WITHOUT RATIONAL TAILS OR BRIDGES HAS AMPLE DUALIZING SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the componentwise reduction, the restriction sequence with the degree identity, the divisibility argument, and the tail/bridge classification of both failure modes are all retained verbatim.

Editorial note. The label lemma-no-rational-bridge-ample-genus-geq-2 names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Ampleness by component degrees. Varieties, Lemma varieties-lemma-ampleness-in-terms-of-degrees-components; Same chapter, Lemmas lemma-genus-gorenstein and lemma-facts-about-nodal-curves. Reduces ampleness to positivity of deg(omega_X|_C) on every irreducible component and handles the irreducible case via the genus identity.
- Divisibility of numerical invariants. Varieties, Lemma varieties-lemma-divisible. Makes [k' : k] divide all numerical invariants of the component C over k' = H^0(C, O_C), forcing the exceptional configurations to be defined over k'.
- The subscheme-complement duality package. Same chapter, Lemmas lemma-closed-subscheme-reduced-gorenstein and lemma-closed-subscheme-nodal-curve. Provides the closure X' of the complement, the restriction sequence 0 -> omega_C -> omega_X|_C -> O_{C cap X'} -> 0, and the nodal structure of the intersection.

Source reference. Original reference 0E3M: sources/stacks-curves/pinned-original-source.tex, source lines 6201–6201. Prerequisite pointer invoked at proof line(s) 6441; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E3H: sources/stacks-curves/pinned-original-source.tex, source lines 5933–5933. Prerequisite pointer invoked at proof line(s) 6439; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0E38: sources/stacks-curves/pinned-original-source.tex, source lines 4794–4794. Prerequisite pointer invoked at proof line(s) 6462; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E34: sources/stacks-curves/pinned-original-source.tex, source lines 1012–1012. Prerequisite pointer invoked at proof line(s) 6459; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E37: sources/stacks-curves/pinned-original-source.tex, source lines 4763–4763. Prerequisite pointer invoked at proof line(s) 6461; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C19: sources/stacks-curves/pinned-original-source.tex, source lines 1744–1744. Prerequisite pointer invoked at proof line(s) 6452; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0B5Y: sources/stacks-chow/tags.txt, source lines 14051–14051. Prerequisite pointer invoked at proof line(s) 6450; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0C6P: sources/stacks-chow/tags.txt, source lines 15304–15304. Prerequisite pointer invoked at proof line(s) 6457; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension 1 and $H^0(X, \mathcal{O}_X) = k$. Assume*

1. *the singularities of X are at-worst-nodal,*
2. *X does not have a rational tail (Example 0E3H),*
3. *X does not have a rational bridge (Example 0E3M),*
4. *the genus g of X is ≥ 2.*

Then ω_X is ample.

Proof. It suffices to show that $\deg(\omega_X|_C) > 0$ for every irreducible component C of X, see Varieties, Lemma 0B5Y. If $X = C$ is irreducible, this follows from $g \geq 2$ and Lemma 0C19. Otherwise, set $k' = H^0(C, \mathcal{O}_C)$. This is a field and a finite extension of k and $[k' : k]$ divides all numerical invariants below associated to C and coherent sheaves on C, see Varieties, Lemma 0C6P. Let $X' \subset X$ be the closure of $X \setminus C$ as in Lemma 0E34. We will use the results of this lemma and of Lemmas 0E37 and 0E38 without further mention. Then we get a short exact sequence

$$0 \to \omega_C \to \omega_X|_C \to \mathcal{O}_{C \cap X'} \to 0$$

See Lemma 0E34. We conclude that

$$\deg(\omega_X|_C) = \deg(C \cap X') + \deg(\omega_C) = \deg(C \cap X') - 2\chi(C, \mathcal{O}_C)$$

Hence, if the lemma is false, then

$$2[k' : k] \geq \deg(C \cap X') + 2 \dim_k H^1(C, \mathcal{O}_C)$$

Since $C \cap X'$ is nonempty and by the divisiblity mentioned above, this can happen only if either

(a) $C \cap X'$ is a single k'-rational point of C and $H^1(C, \mathcal{O}_C) = 0$, and
(b) $C \cap X'$ has degree 2 over k' and $H^1(C, \mathcal{O}_C) = 0$.

The first possibility means C is a rational tail and the second that C is a rational bridge. Since both are excluded the proof is complete. □